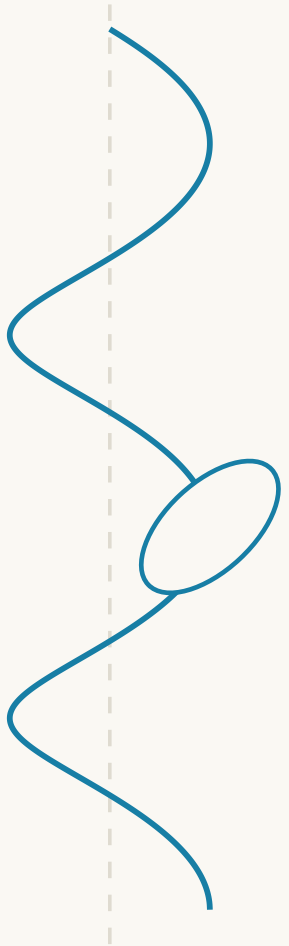
On Leonardo da Vinci's paradox



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joint work with Björn Gebhard and Christian Seis

UZH–ETH Seminar: PDE and Mathematical Physics

October 15, 2026

da Vinci at the ETH



Personal photograph



YouTube Short ▪ Spiraling air bubbles

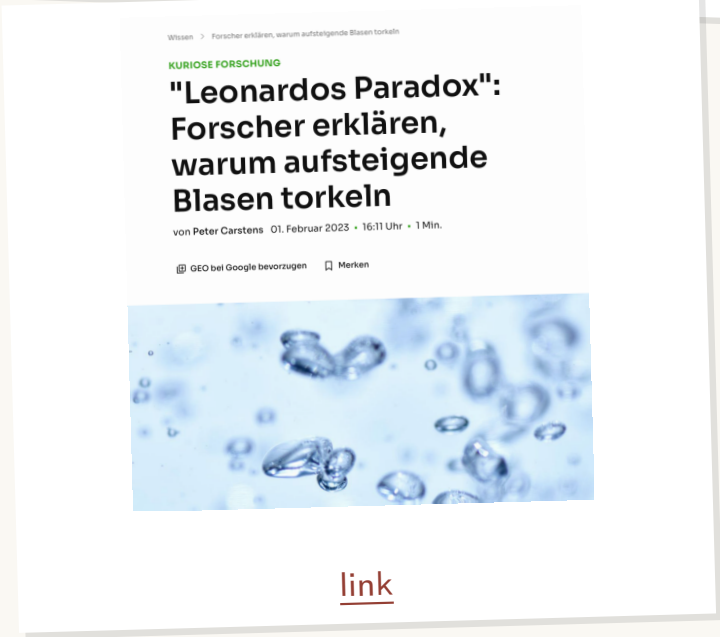
In the press: 2023



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Leonardo da Vinci, 1452–1519

Image source: National Geographic Deutschland.

da Vinci's spiraling bubbles



Leonardo da Vinci, *Codex Leicester*, fol. 25r (detail), c. 1508–10; reproduced in Prosperetti (2004), Fig. 19.

da Vinci's spiraling bubbles



"The air which is submerged together with the water . . . returns to the air, penetrating the water in sinuous movement."

Leonardo da Vinci
Codex Leicester, fol. 25r

Leonardo da Vinci, *Codex Leicester*, fol. 25r; image: Prosperetti (2004), Fig. 18; translation: MacCurdy (1938), I, 112.

Leonardo da Vinci's paradox

"I have proposed to refer to the failure of bubbles to follow a straight path as Leonardo's paradox."

Andrea Prosperetti, "Bubbles," *Phys. Fluids* **16** (2004), 1852–1865, p. 1861.

Today, we prove the existence of steady-spiralling bubbles in an idealised fluid model.



YouTube Short ▪ Sodium–potassium alloy

A bubble in an ambient liquid

We work in $\mathbb{R}^3$.

$\Omega^{\mathrm{out}}(t)$: exterior liquid

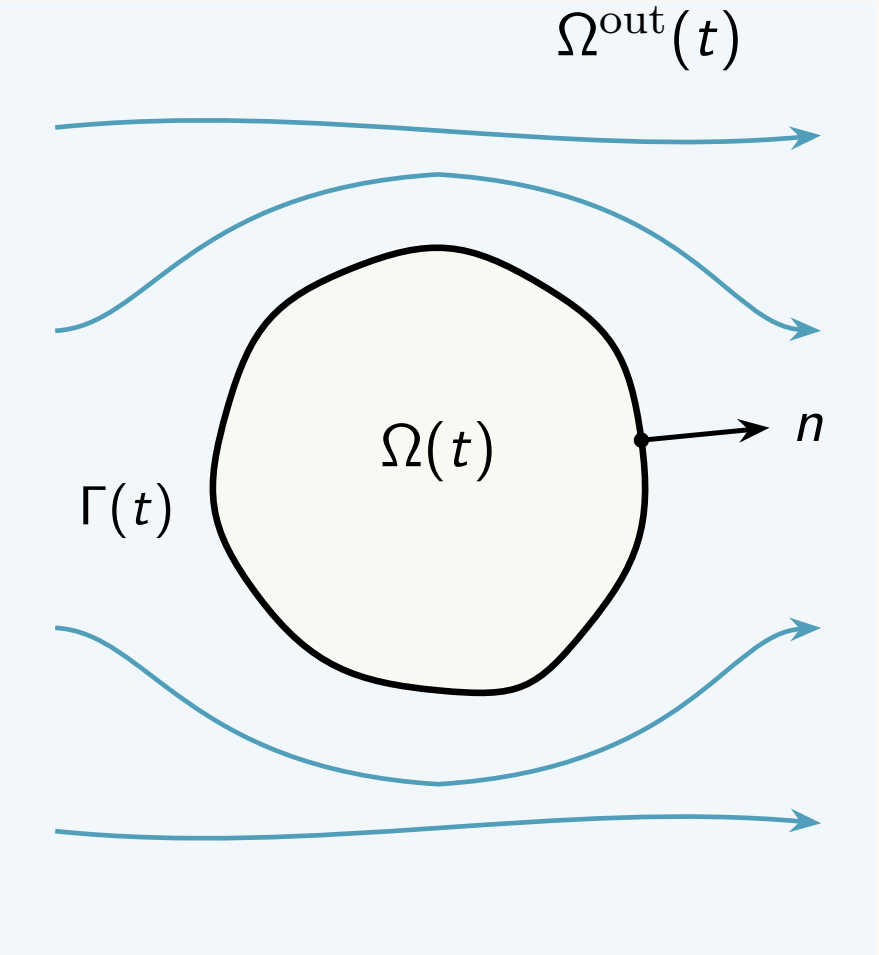
- velocity $u(t,x)$
- pressure $p(t,x)$

$\Omega(t)$: massless bubble

- pressure p_{gas} constant
- zero velocity field

Modeling assumptions

1. Inviscid fluid
2. Incompressible fluid
3. Irrotational exterior flow
4. No gravity



Euler equations

$$\rho\big(\partial_t u + (u\cdot\nabla)u\big) + \nabla p = 0$$

$$\nabla \cdot u = 0$$

incompressible

$$\nabla \times u = 0$$

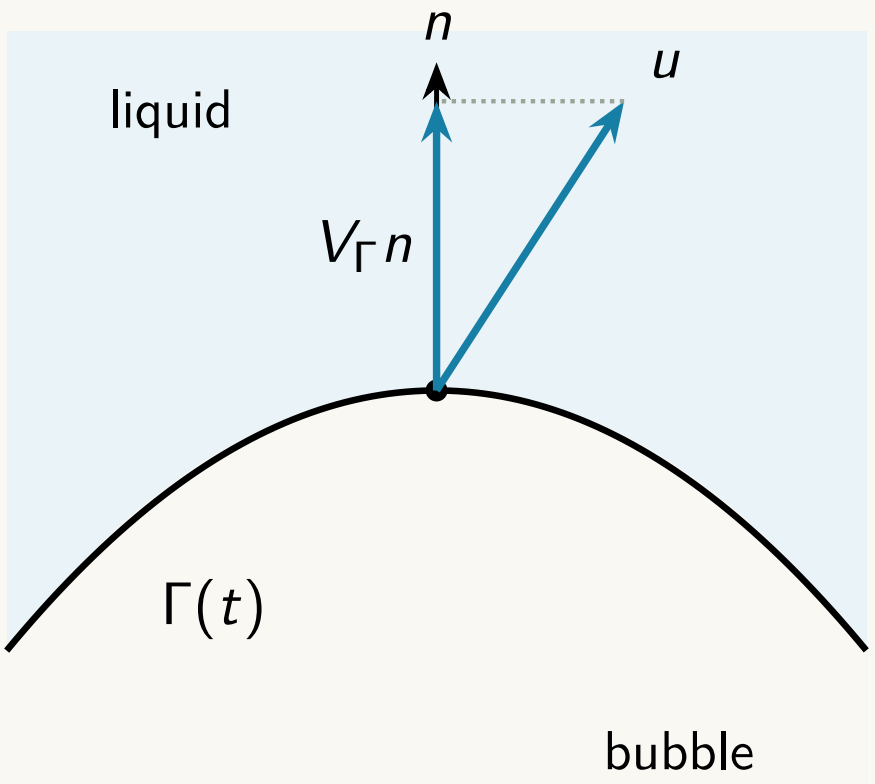
irrotational

Kinematic boundary condition

$$u \cdot n = V_\Gamma$$

The interface and the liquid have the same normal velocity.

Liquid does not cross the free boundary $\Gamma(t)$.



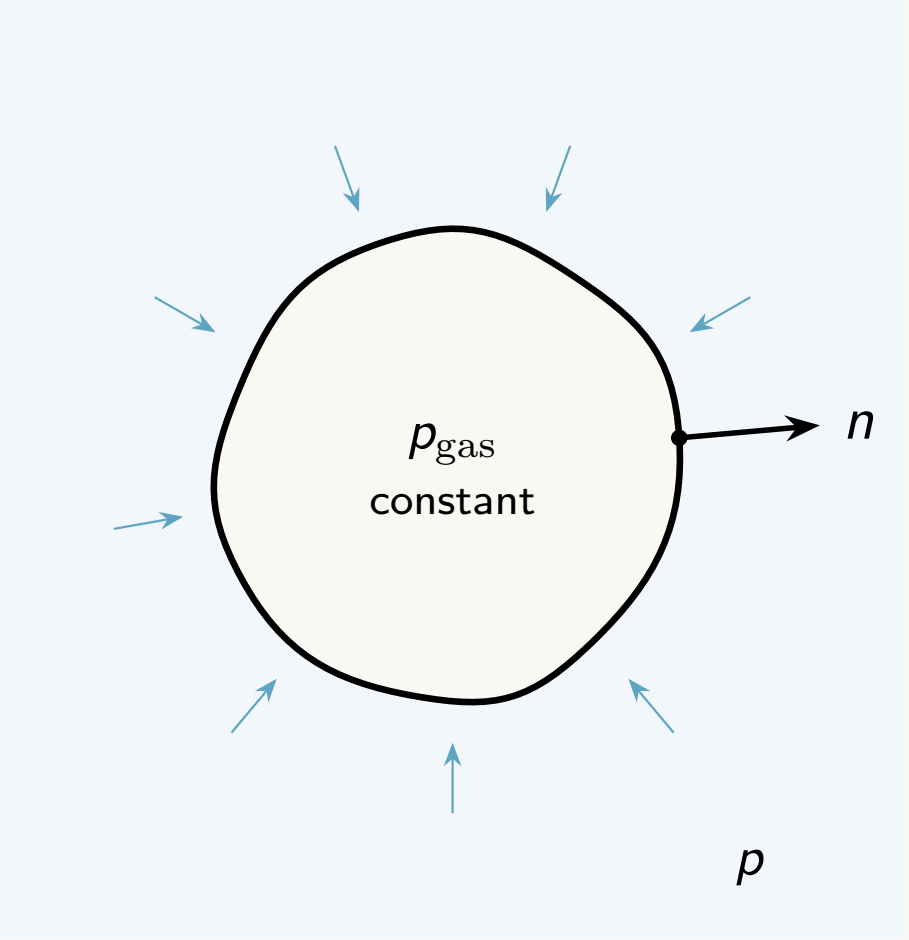
Young–Laplace law

$$p_{\mathrm{gas}} - p = \sigma H$$

$\sigma > 0$: surface-tension coefficient

$H = \mathrm{div}_\Gamma\, n$

Sphere of radius R: $H = 2/R$



We seek a spiralling bubble

$\mathcal{R}_\theta$: rotation through angle θ about e_3.

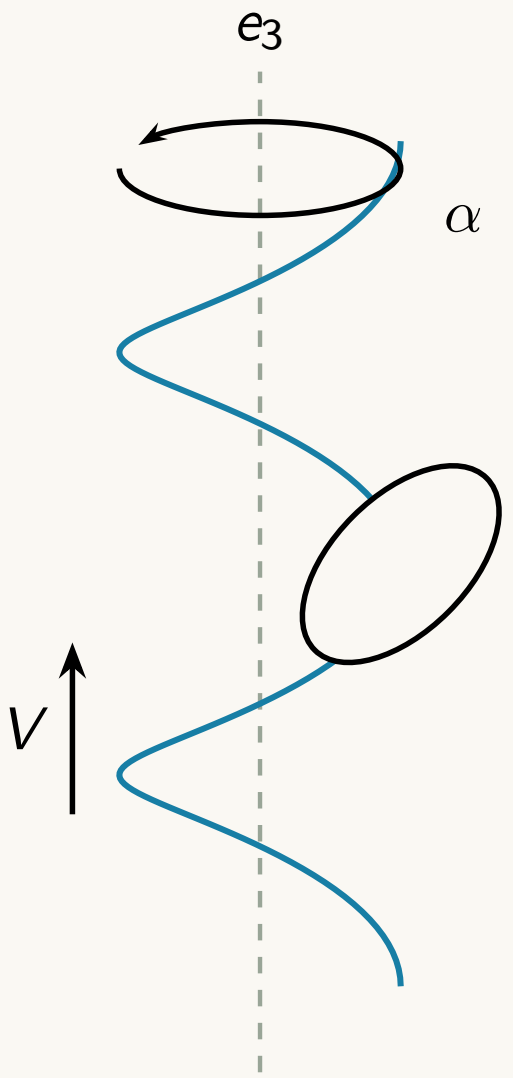
$$x = \mathcal{R}_{\alpha t} y + V t e_3$$

$$\Omega(t) = \mathcal{R}_{\alpha t} \Omega + V t e_3$$

$$u(t, x) = \mathcal{R}_{\alpha t} U(y)$$

V: axial speed; α: angular speed.

The shape and the flow are stationary in the moving frame.



Potential flow

Irrotational flow in a simply connected exterior can be written as

$$u(t,x) = \nabla_x \Phi(t,x)$$

Stationary in the moving frame

$$\Phi(t,x) = \phi(y), \qquad y = \mathcal{R}_{-\alpha t}(x - Vte_3)$$

$$\dot{x} = \alpha e_3 \times (\mathcal{R}_{\alpha t} y) + Ve_3 = \mathcal{R}_{\alpha t} \underbrace{(\alpha e_3 \times y + Ve_3)}_{=: W_{\alpha,V}(y)}$$

Bernoulli and Young–Laplace eliminate the pressure:

$$\sigma H - \rho F_{\alpha,V}(y, \nabla\phi) = c$$

$$F_{\alpha,V}(y,q) = \frac{1}{2}\,|q - Ve_3|^2 + \alpha y \cdot (e_3 \times q)$$

Overdetermined free-boundary value problem

Fix $R, \rho, \sigma > 0$. Find a bounded bubble Ω with simply connected exterior, a potential ϕ, a constant $c \in \mathbb{R}$, and speeds (α, V) such that

$$
\begin{cases}
\Delta\phi = 0 & \text{in } \Omega^{\text{out}}, \\
\partial_n \phi = W_{\alpha,V} \cdot n & \text{on } \Gamma, \\
\sigma H - \rho F_{\alpha,V}(y, \nabla\phi) = c & \text{on } \Gamma, \\
\phi(y) \longrightarrow 0 & \text{as } |y| \longrightarrow \infty, \\
\nabla\phi \in \mathrm{L}^2(\Omega^{\text{out}}), & \\
|\Omega| = \dfrac{4\pi}{3} R^3, & \\
\displaystyle\int_\Omega y_3 \, \mathrm{d}y = 0. &
\end{cases}
$$

Recall $W_{\alpha,V}(y) = \alpha e_3 \times y + V e_3$ and $F_{\alpha,V}(y, q) = \dfrac{1}{2} |q - V e_3|^2 + \alpha y \cdot (e_3 \times q)$.

The Weber number

$$\mathrm{We} := \frac{\rho V^2}{\sigma/R} = \frac{\text{inertial pressure}}{\text{capillary pressure}}$$

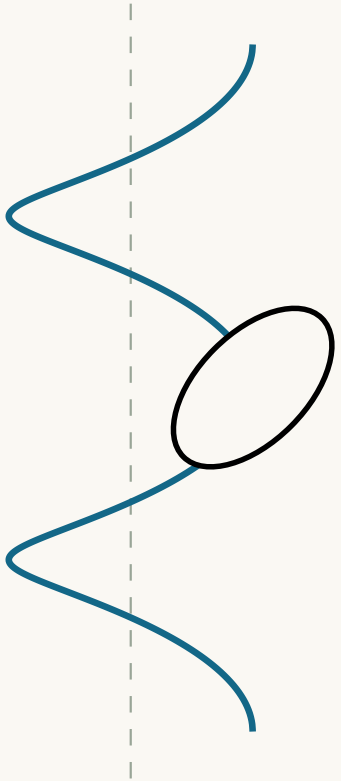
Theorem (Gebhard–N.–Seis '26)

Fix $\rho, \sigma, R > 0$.

For every $0 < \mathrm{We} \ll 1$, the capillary Euler equations admit a near-spherical relative equilibrium whose centroid follows a true helix.

The branch is locally unique near the $(2,1)$ Rayleigh–Lamb frequency, modulo rotations.

$$\alpha \approx \alpha_* = \left(\frac{12\sigma}{\rho R^3}\right)^{1/2}$$



Rigorously confirms predictions by Benjamin (1987) and Meiron (1989).

0. Nondimensionalisation

By rescaling it suffices to consider:

$$R = \rho = \sigma = 1$$

Then

$$\mathrm{We} = V^2 \qquad \text{and} \qquad \alpha_* = 2\sqrt{3}$$

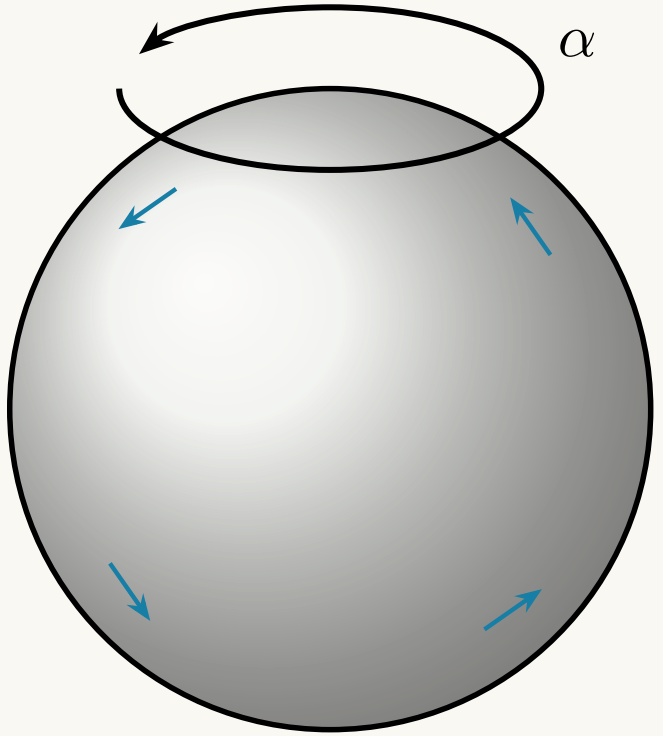
1. The stationary sphere is a solution

$\Gamma = \mathbb{S}^2, \qquad V = 0, \quad \alpha \in \mathbb{R}.$

$n = y \quad \Longrightarrow \quad W_{\alpha,0} \cdot n = 0.$

$\partial_n \phi = 0, \quad \phi \to 0 \quad \Longrightarrow \quad \phi \equiv 0.$

$H - F_{\alpha,0} = 2 = c.$



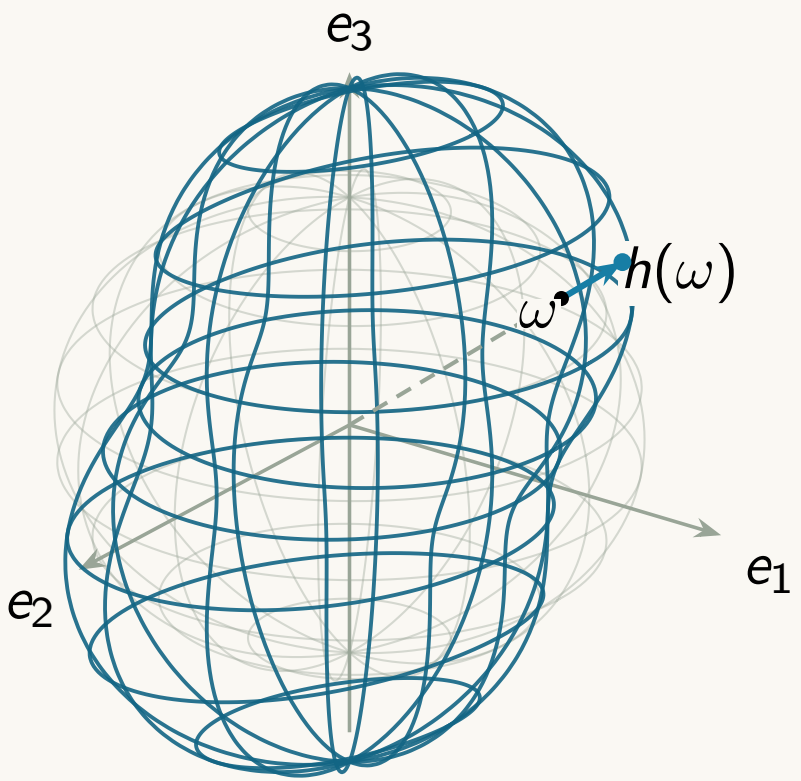
blue arrows: $W_{\alpha,0} = \alpha e_3 \times y$,
tangent to the sphere

2. Spherical graphs

Write $\omega = (X, Y, Z) \in \mathbb{S}^2$. Let $k \geq 4$.
Take $h \in \mathsf{H}^{k+2}(\mathbb{S}^2)$ with $1 + h > 0$.

$$\Psi_h(\omega) = (1 + h(\omega))\omega,$$

$$\Gamma_h = \Psi_h(\mathbb{S}^2).$$



grey: $\mathbb{S}^2$ blue: Γ_h

2. Spherical graphs

The volume and vertical-centroid constraints are

$$\mathcal{V}(h) := \frac{1}{3}\int_{\mathbb{S}^2}(1+h)^3\,\mathrm{d}S - \frac{4\pi}{3}, \qquad \mathcal{C}_3(h) := \frac{1}{4}\int_{\mathbb{S}^2}(1+h)^4 Z\,\mathrm{d}S.$$

$$\mathcal{M}^{k+2} := \{h \in \mathsf{H}^{k+2}(\mathbb{S}^2) :\ \|h\|_{\mathsf{H}^{k+2}} \ll 1,\ \mathcal{V}(h) = \mathcal{C}_3(h) = 0\}.$$

Its tangent space at the sphere is

$$T_0\mathcal{M}^{k+2} = \left\{\xi \in \mathsf{H}^{k+2}(\mathbb{S}^2) : \int_{\mathbb{S}^2}\xi\,\mathrm{d}S = 0,\ \int_{\mathbb{S}^2}\xi Z\,\mathrm{d}S = 0\right\} =: \mathsf{H}^{k+2}_{\mathrm{lin}}\,.$$

The implicit function theorem gives the local chart

$$\Theta : \mathcal{U} \subset T_0\mathcal{M}^{k+2} \longrightarrow \mathcal{M}^{k+2}, \qquad h = \Theta(\xi) = \xi + c(\xi) + d(\xi)Z.$$

2. Spherical graphs

For each h, solve the exterior Neumann problem for ϕ_h and set

$$\mathcal{B}(h, \alpha, V) = \Pi_h(H_h \circ \Psi_h - F_{\alpha,V}(y, \nabla\phi_h) \circ \Psi_h)$$

Projection to remove the unknown Bernoulli constant c.

$$\Pi_h f = f - \frac{\int_{\mathbb{S}^2} f(1+h)^2 \,\mathrm{d}S}{\int_{\mathbb{S}^2}(1+h)^2 \,\mathrm{d}S}$$

Bernoulli map

$$\mathcal{B} : \mathsf{H}^{k+2}_{\mathrm{lin}} \times \mathbb{R}^2 \longrightarrow \mathsf{H}^{k}_{\mathrm{lin}}$$

Goal: find (h, α, V) such that $\mathcal{B}(h, \alpha, V) = 0$

3. Linearisation at the sphere

Spherical harmonics Y_ℓ^m:

$$-\Delta_{\mathbb{S}^2} Y_\ell^m = \ell(\ell+1) Y_\ell^m, \qquad \ell \in \mathbb{N}_0, \quad |m| \le \ell.$$

Lemma

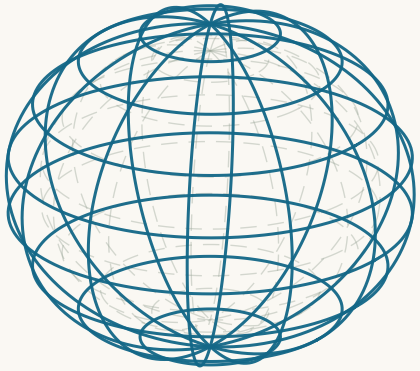
$$D_h \mathcal{B}(0,\alpha,0) Y_\ell^m = \begin{cases} \lambda_{\ell m}(\alpha) Y_\ell^m, & \ell \ge 1, \\ 0, & \ell = 0. \end{cases}$$

$$\lambda_{\ell m}(\alpha) = \underbrace{\ell(\ell+1)-2}_{\substack{\text{mean curvature} \\ -(\Delta_{\mathbb{S}^2}+2)}} - \underbrace{\alpha^2 \frac{m^2}{\ell+1}}_{\text{fluid rotation}}$$

4. Kernel I: constraints remove two modes

$Y_0^0 = 1, \qquad Y_1^0 = Z$ are kernel modes for every α.

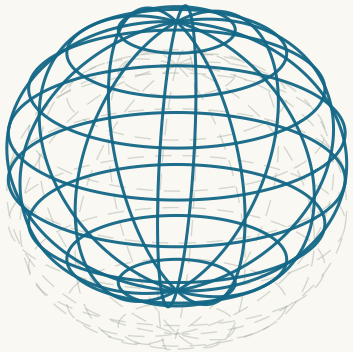
1



fixed volume: $\int_{\mathbb{S}^2} \xi \, \mathrm{d}S = 0$

removes the radial direction

Z



fixed vertical centroid: $\int_{\mathbb{S}^2} \xi Z \, \mathrm{d}S = 0$

removes vertical translation

5. Axisymmetric bubbles

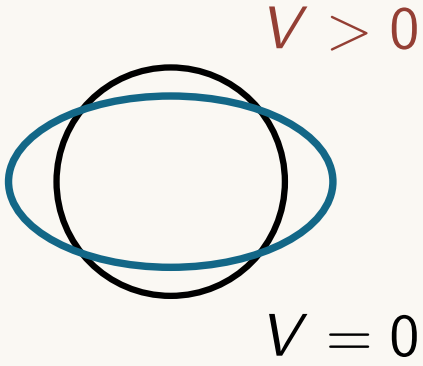
Restrict to $h = h(Z)$. Then only spherical harmonics with $m = 0$ are admissible. Linearisation is invertible:

$$D_h\mathcal{B}(0, \alpha, 0)\, Y_\ell^0 = (\ell(\ell+1) - 2)\, Y_\ell^0, \qquad \ell \geq 2$$

Theorem (Meyer–N.–Seis, 2025)

For $0 < V \ll 1$, there is a locally unique smooth translating bubble of volume $4\pi/3$, axisymmetric and symmetric about $Z = 0$.

$$h_{0,V}^{\mathrm{ax}} = -\frac{3V^2}{16}\,\frac{3Z^2 - 1}{2} + O_{\mathsf{H}^{k+2}}(V^4).$$



In particular, $\mathcal{B}(h_{0,V}^{\mathrm{ax}}, \alpha, V) = 0$ for every α.

The oblate deformation was predicted by Moore (1965).

6. Kernel II

Consider now $\alpha = \alpha_* = 2\sqrt{3}$.

$$\lambda_{21}(\alpha_*) = 0$$

degree 2, order 1

$$K_1 = \text{span}\{XZ, YZ\}$$

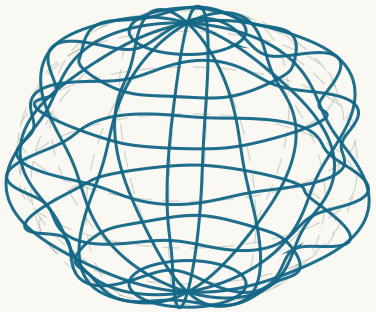


desired bifurcation direction

$$\lambda_{76}(\alpha_*) = 0$$

degree 7, order 6

$$K_6 = \text{span}\left\{ \begin{matrix} Z\,\text{Re}(X+iY)^6, \\ Z\,\text{Im}(X+iY)^6 \end{matrix} \right\}$$



unwanted resonances

grey: sphere blue: kernel direction

7. Bernoulli as the gradient of the action

$$\mathcal{E}(\Omega, \alpha, V) = |\partial\Omega| - \frac{1}{2}\int_{\Omega^{\mathrm{out}}} \left|\nabla \phi_{\Omega}^{\alpha, V}\right|^2 \, \mathrm{d}y$$

On the constrained chart $h = \Theta(\xi)$, set

$$\mathcal{J}(\xi, \alpha, V) := \mathcal{E}(\Omega_{\Theta(\xi)}, \alpha, V), \qquad \mathcal{F}(\xi, \alpha, V) := \nabla_{\xi}^{\mathrm{L}^2} \mathcal{J}(\xi, \alpha, V).$$

Lemma

We have

$$\mathcal{B}(\Theta(\xi), \alpha, V) = 0 \quad \Longleftrightarrow \quad \mathcal{F}(\xi, \alpha, V) = 0.$$

Moreover, their kernels and invertibility properties are equivalent.

8. Lyapunov–Schmidt reduction

P_j: L^2-projection onto K_j, $j = 1, 6$.

$Q := \mathrm{Id} - P_1 - P_6$.

$x = a + u, \qquad a \in K_1, \quad u \in K_6$.

Range equation

$$Q\mathcal{F}(x + w(x, \alpha, V), \alpha, V) = 0, \qquad w(x, \alpha, V) \in (K_1 \oplus K_6)^{\perp}.$$

Uniquely solvable for $w = w(x, \alpha, V)$ by the implicit function theorem.

Kernel equations

$$\Phi_j(a, u, \alpha, V) \;:=\; P_j\mathcal{F}(x + w(x, \alpha, V), \alpha, V), \qquad j \;=\; 1, 6.$$

$$\Phi_1 = 0, \qquad \Phi_6 = 0$$

Four scalar equations remain.

9. Symmetry and equivariance

Recall: R_θ is the rotation by θ about e_3. Define

$$(\mathcal{U}_\theta h)(\omega) := h(R_{-\theta}\omega), \qquad \omega = (X, Y, Z) \in \mathbb{S}^2.$$

We study this action on $K_1 = \mathrm{span}\{XZ, YZ\}$:

$$\mathcal{U}_\theta(XZ) = \cos\theta\, XZ + \sin\theta\, YZ,$$
$$\mathcal{U}_\theta(YZ) = -\sin\theta\, XZ + \cos\theta\, YZ.$$

On K_j, $j = 1, 6$,

$$\mathcal{U}_\theta|_{K_j} = \cos(j\theta)\ \mathrm{Id} + \sin(j\theta)\,\mathrm{J}, \qquad \mathrm{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

10. $4 \mapsto 3$

Consider the reduced action

$$\widehat{\mathcal{J}}(a,u) := \mathcal{J}(a+u+w(a+u,\alpha,V),\alpha,V)$$

$$(\Phi_1,\Phi_6) = \nabla_{(a,u)}\widehat{\mathcal{J}}, \qquad \widehat{\mathcal{J}}(\mathcal{U}_\theta a,\mathcal{U}_\theta u) = \widehat{\mathcal{J}}(a,u).$$

$$0 \;=\; \frac{\mathrm{d}}{\mathrm{d}\theta}\widehat{\mathcal{J}}(\mathcal{U}_\theta a,\mathcal{U}_\theta u)\Big|_{\theta=0} \;=\; \langle \Phi_1, \mathrm{J}a\rangle + 6\langle \Phi_6, \mathrm{J}u\rangle.$$

For $a \neq 0$, the vectors a and $\mathrm{J}a$ span K_1, hence

$$\Phi_1 = \Phi_6 = 0 \quad\Longleftrightarrow\quad \langle \Phi_1, a\rangle = 0, \quad \Phi_6 = 0.$$

11. Moving off the sphere lifts the resonance

At $(a, u) = (0, 0)$,

$$D_{(a,u)} \begin{pmatrix} \Phi_1 \\ \Phi_6 \end{pmatrix} (0, 0, \alpha, V) = \begin{pmatrix} \mu_1(\alpha, V) I_{K_1} & 0 \\ 0 & \mu_6(\alpha, V) I_{K_6} \end{pmatrix}$$

11. Moving off the sphere lifts the resonance

We need to expand

$$D_h\mathcal{B}(h^{\mathrm{ax}}_{0,V},\alpha,V) = L_0^\alpha + VL_1^{\alpha_*} + V^2L_2^{\alpha_*} + O(V^3)$$

uniformly for $|\alpha-\alpha_*| \lesssim V^2$.

$L_0^\alpha Y_\ell^m = \lambda_{\ell m}(\alpha) Y_\ell^m$	diagonal in spherical harmonics
$L_1^{\alpha_*}$	preserves m and couples ℓ only to $\ell \pm 1$
$L_2^{\alpha_*}$	only its diagonal matrix elements are needed

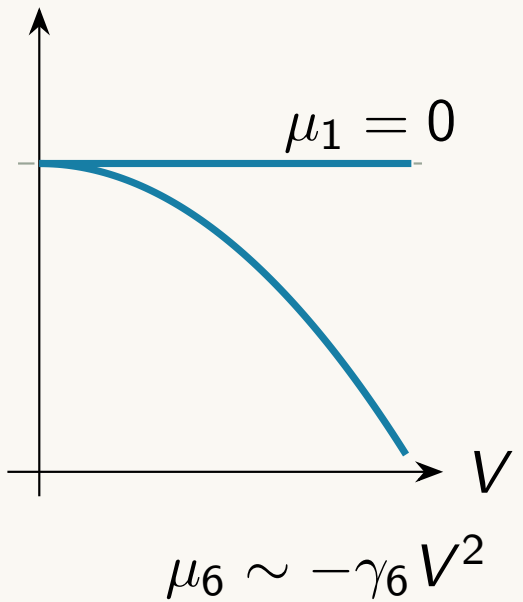
11. Moving off the sphere lifts the resonance

Uniformly for $|\alpha - \alpha_*| \lesssim V^2$,

$$\mu_1(\alpha, V) = \lambda_{21}(\alpha) - \frac{867}{980} V^2 + O(V^3),$$

$$\mu_6(\alpha, V) = \lambda_{76}(\alpha) - \frac{4174947}{284240} V^2 + O(V^3),$$

$$\partial_\alpha \mu_1(\alpha_*, 0) = \partial_\alpha \lambda_{21}(\alpha_*) = -\frac{2\alpha_*}{3} \neq 0.$$



Implicit function theorem to choose $\alpha_1(V)$ with

$$\mu_1(\alpha_1(V), V) = 0, \qquad \alpha_1(V) = 2\sqrt{3} - \frac{2601}{1960} \frac{V^2}{2\sqrt{3}} + O(V^3).$$

The resonance is lifted, but inversion costs V^{-2}:

$$\mu_6(\alpha_1(V), V) = -\gamma_6 V^2 + O(V^3), \qquad \gamma_6 = \frac{38228049}{13927760} > 0.$$

12. Nonlinearities

Consider now $a = sXZ$ and $\alpha = \alpha_1(V)$.

$$\Phi_1(sXZ, u) = \mu_1(\alpha, V)sXZ + O\Big(|s|^3 + |s|^5 \|u\| + |s| \|u\|^2\Big),$$

$$\Phi_6(sXZ, u) = \mu_6(\alpha, V)u + O\Big(s^6 + s^2 \|u\| + \|u\|^2\Big).$$

Sixth-order forcing compensates for the V^{-2} loss:

$$\Big\|\mu_6(\alpha_1(V), V)^{-1}\Phi_6(sXZ, 0)\Big\| \lesssim \frac{|s|^6}{V^2}.$$

12. Nonlinearities

Write the coefficient pairs in K_1 and K_6 as complex numbers z and q. A rotation through θ acts as

$$z \quad \longmapsto \quad e^{i\theta} z, \qquad q \quad \longmapsto \quad e^{6i\theta} q.$$

$$\Phi_6(e^{i\theta} z, e^{6i\theta} q) = e^{6i\theta} \Phi_6(z, q)$$

Set $q = 0$. A Taylor monomial $z^p \bar{z}^r$ can occur only if

$$p - r = 6 \quad \Longrightarrow \quad p + r \geq 6.$$

Hence,

$$\Phi_6(z, 0) = O(|z|^6)$$

13. Blow-up

We rescale as

$$s = Vt, \qquad u = V^2 U, \qquad \alpha = \alpha_1(V) + V^2\beta.$$

$$\widetilde{\Phi}_1 = \frac{\langle \Phi_1, XZ \rangle}{V^3 t \, \|XZ\|^2_{\mathrm{L}^2}}, \qquad \widetilde{\Phi}_6 = \frac{\Phi_6}{V^4}$$

Both Φ_j are evaluated at $(VtXZ, V^2U, \alpha_1(V) + V^2\beta, V)$.

Equivariance makes the rescaled equations smooth through $Vt = 0$.

13. Blow-up

At $(t, U, \beta, V) = (0, 0, 0, 0)$,

$$D_{(\beta,U)}(\widetilde{\Phi}_1, \widetilde{\Phi}_6) = \begin{pmatrix} -2\alpha_*/3 & 0 \\ 0 & -\gamma I_{K_6} \end{pmatrix}.$$

Implicit function theorem gives $\beta_*(t, V)$ and $U_*(t, V)$ so that

$$\widetilde{\Phi}_j(t, U_*(t, V), \beta_*(t, V), V) = 0, \qquad j = 1, 6.$$

Undoing the rescaling gives, for $0 < V < V_0$ and $|s| \le \eta V$, $u(s, V)$ and $\alpha(s, V)$ with

$$\|u(s, V)\| \lesssim \frac{|s|^6}{V^2}, \qquad \alpha(s, V) = \alpha_1(V) + O(s^2).$$

$$\xi(s, V) = sXZ + u(s, V) + w(sXZ + u(s, V), \alpha(s, V), V)$$

$$\mathcal{F}(\xi(s, V), \alpha(s, V), V) = 0$$

14. The centroid follows a true helix

Calculating the asymptotic expansion yields

$$\alpha(s, V) = 2\sqrt{3} - \frac{2601}{1960}\frac{V^2}{2\sqrt{3}} + O(s^2 + V^3).$$

$$\begin{aligned} h(s, V) = & -\frac{3V^2}{16}\frac{3Z^2 - 1}{2} + sXZ - \frac{9}{20\sqrt{3}}VsY \\ & - \frac{5\sqrt{3}}{7}VsY\left(Z^2 - \frac{1}{5}\right) + O_{\mathsf{H}^{k+2}}\left(s^2 + |s|\, V^2 + V^4\right). \end{aligned}$$

14. The centroid follows a true spiral

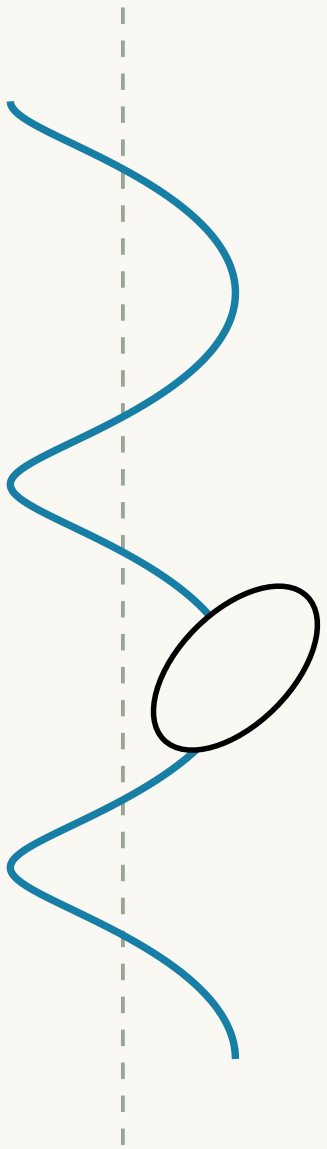
$$\bar{y}(s,V) = \frac{1}{|\Omega(s,V)|}\int_{\Omega(s,V)} y\,\mathrm{d}y.$$

$$\pi_{e_3^\perp}\bar{y}(s,V) = -\frac{9}{20\sqrt{3}}Vs\,e_2 + O\Big(s^2 + |s|\,V^2\Big).$$

$$0 < |s| \le \eta V \quad \Longrightarrow \quad \Big|\pi_{e_3^\perp}\bar{y}(s,V)\Big| > 0.$$

$$\bar{x}(t) = \mathcal{R}_{\alpha t}\bar{y}(s,V) + Vte_3.$$

Hence, the centroid follows a genuine helix.



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